Lesson plan · for the teacher

Inside a language model

AI · Level 3 · Lesson 4

Time	Materials
40 min	Board and coloured chalk; Three sentences where “it” could mean two things; A page from your textbook (for the grounding demonstration)

Learning goals

- Describe the loop: tokens become numbers, attention gathers context, the model scores every possible next token, picks one and repeats.
- Explain the training stages: pretraining on huge amounts of text, then fine-tuning with examples and human feedback.
- Explain why a language model can be confidently wrong, and how grounding and checking help.

Activity: Who is “it”?

1. Warm-up (5 min): Write “The bus didn’t stop at the school because it was…” on the board. Pupils call out endings. Ask: what does “it” mean in each ending?
2. Who is “it”? (12 min): Write three sentences, such as “The goat ate the maize because it was hungry.” In pairs, pupils copy them and draw arrows from “it” to the words that tell them what it means. Compare the arrows on the board.
3. Change one word (6 min): Change “hungry” to “sweet”. Pupils redraw the arrows: now “it” is the maize. Ask: which single word changed the meaning?
4. The loop (9 min): Draw the loop: tokens, numbers, attention, scores, pick, repeat. Explain that attention does what the arrows did: the newest word weighs the earlier ones. Play “human language model”: the class scores three possible next words for a sentence and picks one.
5. Wrap-up (8 min): Show a chatbot answer with a wrong local fact (prepared in advance) and a textbook page. Ask: how would grounding help, and what would we still check? Then give the exit ticket.

Prepare three “it” sentences in the pupils’ language, a printed chatbot answer that contains one wrong local fact, and the textbook page that corrects it.

Discussion

- How did you know what “it” meant? Which words did you look back at?
- Why might a chatbot get a fact about our town wrong?
- What source would you give a chatbot before trusting its answer?

Exit ticket

- Write the loop in five words, and one reason a language model can be confidently wrong.

Try it at school tomorrow

Play “Who is ‘it’?”: write three sentences on the board where “it” could mean two things, such as “The goat ate the maize because it was hungry.” Pupils draw arrows from “it” to the words that tell them what it means. Then change one word (“hungry” to “sweet”) and redraw the arrows. Explain that a language model’s attention does the same job: it weighs earlier words before it writes the next one.

Practice questions · for you

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1 **Why can a language model sound sure and still be wrong?**

- ◯ It copied a wrong website
- ◯ It skipped the fine-tuning stage
- ◯ It is lying on purpose
- ◯ It predicts likely text, not facts

2 **Put one round of writing in order.**

- ☐ Score every possible next token
- ☐ Layers of neurons work on the numbers
- ☐ Pick one and add it to the text
- ☐ Turn the tokens so far into lists of numbers
- ☐ Attention: the newest token weighs the earlier ones

3 **If raters always prefer a full answer to "I don't know", fine-tuning teaches the model to guess when it doesn't know.**

- ◯ True
- ◯ False

4 **Match each stage to what happens in it.**

Pretraining	• • • •	People rate which of two answers is better
Fine-tuning with examples	• • • •	Predict the next token across huge amounts of text
Human feedback	• • • •	Give it a trusted source to answer from
Grounding	• • • •	Learn from good answers that people wrote

✎ **Where would grounding help most in your teaching? Which source would you give a chatbot, and what would you still check yourself?**

Check your answers

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1 **It predicts likely text, not facts**

The loop scores likely next tokens, and nothing in it checks the truth, so a smooth, sure sentence can be false.

2 **Top to bottom: 4, 3, 5, 1, 2**

Words must become numbers before anything can be computed; attention gathers the context, the layers work on it, and only then are the next tokens scored and one is picked.

3 **True**

Fine-tuning nudges the model towards answers people rate higher.

4 **Pretraining — Predict the next token across huge amounts of text**
Fine-tuning with examples — Learn from good answers that people wrote
Human feedback — People rate which of two answers is better
Grounding — Give it a trusted source to answer from

Pretraining makes the model fluent; fine-tuning with examples and people's ratings makes it helpful; grounding happens when you use it, by giving it a source.